

CHAPTER 6

Problems with the Error

The standard assumption about the error term ϵ is that it is independent and identically distributed (i.i.d.) from case to case. That is, $\text{var } \epsilon = \sigma^2 I$. Furthermore, we also assume that the errors are normally distributed in order to carry out the usual statistical inference. We have seen that these assumptions can often be violated and we must then consider alternatives. When the errors are not i.i.d., we consider the use of *generalized least squares* (GLS). When the errors are independent, but not identically distributed, we can use *weighted least squares* (WLS), which is a special case of GLS. Sometimes, we have a good idea how large the error should be, but the residuals may be much larger than we expect. This is evidence of a *lack of fit*. When the errors are not normally distributed, we can use *robust regression*.

6.1 Generalized Least Squares

Until now we have assumed that $\text{var } \epsilon = \sigma^2 I$, but sometimes the errors have nonconstant variance or are correlated. Suppose instead that $\text{var } \epsilon = \sigma^2 \Sigma$ where σ^2 is unknown but Σ is known—in other words, we know the correlation and relative variance between the errors, but we do not know the absolute scale. Right now, it might seem redundant to distinguish between σ and Σ , but we will see how this will be useful later.

We can write $\Sigma = SS^T$, where S is a triangular matrix using the Choleski decomposition. Now we can transform the regression model as follows:

$$\begin{aligned} y &= X\beta + \epsilon \\ S^{-1}y &= S^{-1}X\beta + S^{-1}\epsilon \\ y' &= X'\beta + \epsilon' \end{aligned}$$

Now we find that:

$$\text{var } \epsilon' = \text{var } (S^{-1}\epsilon) = S^{-1}(\text{var } \epsilon)S^{-T} = S^{-1}\sigma^2 SS^T S^{-T} = \sigma^2 I$$

So we can reduce GLS to ordinary least squares (OLS) by a regression of $y' = S^{-1}y$ on $S^{-1}X$ which has error ϵ' that is i.i.d. So we simply reduce the problem to one that we have already solved. In this transformed model, the sum of squares is:

$$(S^{-1}y - S^{-1}X\beta)^T (S^{-1}y - S^{-1}X\beta) = (y - X\beta)^T S^{-T} S^{-1} (y - X\beta) = (y - X\beta)^T \Sigma^{-1} (y - X\beta)$$

which is minimized by:

$$\hat{\beta} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y$$

We find that:

$$\text{var } \hat{\beta} = (X^T \Sigma^{-1} X)^{-1} \sigma^2$$

Since $\varepsilon' = S^{-1}\varepsilon$, diagnostics should be applied to the residuals, $S^{-1}\hat{\varepsilon}$. If we have the right Σ , then these should be approximately i.i.d.

The main problem in applying GLS in practice is that Σ may not be known and we may have to estimate it. To illustrate this we will use a built-in R dataset known as Longley's regression data. Our response is the number people employed, yearly from 1947 to 1962 and the predictors are gross national product (GNP) and population 14 years of age and over. The data originally appeared in Longley (1967).

Fit a linear model:

```
> data (longley)
> g <- lm (Employed ~ GNP + Population, longley)
> summary (g, cor=T)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  88.9388    13.7850    6.45 2.2e-05
GNP           0.0632     0.0106    5.93 5.0e-05
Population   -0.4097     0.1521   -2.69 0.018
Residual standard error: 0.546 on 13 degrees of freedom
Multiple R-Squared: 0.979, Adjusted R-squared: 0.976
F-statistic: 304 on 2 and 13 DF, p-value: 1.22e-11
Correlation of Coefficients:
              (Intercept) GNP
GNP           0.98
Population   -1.00      -0.99
```

The correlation between the coefficients for GNP and Population is strongly negative while the correlation between the corresponding variables:

```
> cor (longley$GNP, longley$Pop)
[1] 0.99109
```

is strongly positive.

In data collected over time such as this, successive errors could be correlated. The simplest way to model this is the autoregressive form:

$$\varepsilon_{i+1} = \rho_i + \delta_i$$

where $\delta_i \sim N(0, \tau^2)$. We can estimate this correlation ρ by:

```
> cor (residuals (g) [-1], residuals (g) [-16])
[1] 0.31041
```

Under this assumption $\Sigma_{ij} = \rho^{i-j}$. For simplicity, let's assume we know that $\rho = 0.31041$. We now construct the Σ matrix and compute the GLS estimate of β along with its standard errors. The calculation is for demonstration purposes only:

```
> x <- model.matrix (g)
> Sigma <- diag (16)
> Sigma <- 0 . 31041^ abs (row(Sigma) -col(Sigma))
> Sigi <- solve (Sigma)
> xtxi <- solve (t (x) %*% Sigi %*% x)
> (beta <- solve (t (x) %*% Sigi %*% x, t (x) %*%
Sigi %*% longley$Empl))
      [,1]
(Intercept) 94.89889
GNP 0.06739
Population -0.47427
> res <- longley$Empl - x %*% beta
> (sig <- sqrt ( (t (res) %*% Sigi %*% res)/g$sdf))
      [,1]
[1,] 0.5424432
> sqrt (diag (xtxi))*sig
[1] 13.94477260 0.01070339 0.15338547
```

Compare with the model output above where the errors are assumed to be uncorrelated. Another way to get the same result is to regress $S^{-1}y$ on $S^{-1}x$ as we demonstrate here:

```
> sm <- chol (Sigma)
> smi <- solve (t (sm))
> sx <- smi %*% x
> sy <- smi %*% longley$Empl
> summary(lm(sy ~ sx -1))
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
sx (Intercept)   94.8989    13.9448     6.81 1.3e-05
sxGNP             0.0674     0.0107     6.30 2.8e-05
sxPopulation    -0.4743     0.1534    -3.09 0.0086
Residual standard error: 0.542 on 13 degrees of freedom
```

In practice, we would not know that the $\rho=0.31$ and we will need to estimate it from the data. Our initial estimate is 0.31, but once we fit our GLS model we would need to reestimate it as:

```
> cor (res [ -1], res [-16])
[1] 0.35642
```

and then recompute the model again with $\rho=0.35642$. This process would be iterated until convergence. This is cumbersome. A more convenient approach may be found in the nlme package of Pinheiro and Bates (2000), which contains a GLS fitting function. We can use it to fit this model:

```

> library (nlme)
> g <- gls (Employed ~ GNP + Population,
  correlation=corAR1(form= ~ Year), data=longley)
> summary (g)
Correlation Structure: AR(1)
  Formula: ~ Year
  Parameter estimate (s):
    Phi
    0.64417
Coefficients:
              Value Std.Error t-value p-value
(Intercept) 101.858 14.1989   7.1736 <.0001
GNP           0.072  0.0106   6.7955 <.0001
Population   -0.549  0.1541-3.5588  0.0035
Residual standard error: 0.68921
Degrees of freedom: 16 total; 13 residual

```

We see that the estimated value of ρ is 0.64. However, if we check the confidence intervals for this:

```

> intervals (g)
Approximate 95% confidence intervals
Coefficients:
              lower      est.      upper
(Intercept) 71.183204 101.858133 132.533061
GNP           0.049159  0.072071  0.094983
Population   -0.881491 -0.548513 -0.215536
Correlation structure:
              lower      est.      upper
Phi -0.44335 0.64417 0.96451
Residual standard error:
              lower      est.      upper
0.24772 0.68921 1.91748

```

we see from the interval, $(-0.44, 0.96)$, that it is not significantly different from zero. So there is no evidence of serial correlation.

6.2 Weighted Least Squares

Sometimes the errors are uncorrelated, but have unequal variance where the form of the inequality is known. When Σ is diagonal, the errors are uncorrelated but do not necessarily have equal variance. WLS can be used in this situation. We can write $\Sigma = \text{diag}(1/w_1, \dots, 1/w_n)$, where the w_i are the weights so $S = \text{diag}(\sqrt{1/w_1}, \dots, \sqrt{1/w_n})$

So we can regress $\sqrt{w_i}y_i$ on $\sqrt{w_i}x_i$ (although the column of ones in the X -matrix needs to be replaced with $\sqrt{w_i}$). Cases with low variability should get a high weight, high variability a low weight. Some examples:

1. Errors proportional to a predictor: $\text{var}(\epsilon_i) \propto x_i$ suggests $w_i = x_i^{-1}$.
2. When the Y_i are the averages of n_i observations, then $\text{var } y_i = \text{var } \epsilon_i = \sigma^2/n_i$, which suggests $w_i = n_i$. Responses that are averages arise quite commonly, but take care that the variance in the response really is proportional to the group size. For example, consider the life expectancy for different countries. At first glance, one might consider setting the weights equal to the populations of the countries, but notice that there are many other sources of variation in life expectancy that would dwarf the population size effect. Setting $w_i = n_i$ is only likely to be sensible for small n_i .

When weights are used, the residuals must be modified. Use $\sqrt{w_i}\hat{\epsilon}_i$ for diagnostics.

Elections for the French presidency proceed in two rounds. In 1981, there were 10 candidates in the first round. The top two candidates then went on to the second round, which was won by François Mitterand over Valéry Giscard-d'Estaing. The losers in the first round can gain political favors by urging their supporters to vote for one of the two finalists. Since voting is private, we cannot know how these votes were transferred; we might hope to infer from the published vote totals how this might have happened. Anderson and Loynes (1987) published data on these vote totals in every fourth department of France:

```
> data (fpe)
> fpe
```

	EI	A	B	C	D	E	F	G	H	J	K	A2	B2
Ain	260	51	64	36	23	9	5	4	4	3	3	105	114
Alpes	75	14	17	9	9	3	1	2	1	1	1	32	31
...													

A and B stand for Mitterand's and Giscard's votes in the first round, respectively, while A2 and B2 represent their votes in the second round. C–K are the first round votes of the other candidates while EI is *electeur inscrits* or registered voters. All numbers are in thousands. The total number of voters in the second round was greater than the first—we can compute the difference as N.

We will treat this group effectively as another first round candidate (we could reasonably handle this differently). Now we can represent the transfer of votes as:

$$A2 = \beta_A A + \beta_B B + \beta_C C + \beta_D D + \beta_E E + \beta_F F + \beta_G G + \beta_H H + \beta_J J + \beta_K K + \beta_N N$$

where β_i represents the proportion of votes transferred from candidate i to Mitterand in the second round. Now we would expect these transfer proportions to vary somewhat between departments, so if we treat the above as a regression equation, there will be some error from department to department. The error will have a variance in proportion to the number of voters because it will be like a variance of a sum rather than a mean. Since the weights should be inversely proportional to the variance, this suggests that the weights should be set to $1/EI$. Notice also that the equation has no intercept, hence the -1 in the model formula. We fit the appropriate model:

```
> g <- lm (A2 ~ A+B+C+D+E+F+G+H+J+K+N-1, fpe,
  weights=1/EI)
> coef (g)
      A      B      C      D      E      F      G
1.06713 -0.10505 0.24596 0.92619 0.24940 0.75511 1.97221
      H      J      K      N
-0.56622 0.61164 1.21066 0.52935
```

Note that the weights do matter—see what happens when we leave them out:

```
> lm (A2 ~ A+B+C+D+E+F+G+H+J+K+N-1, fpe) $coef
      A      B      C      D      E      F      G
1.07515 -0.12456 0.25745 0.90454 0.67068 0.78253 2.16566
      H      J      K      N
-0.85429 0.14442 0.51813 0.55827
```

which causes substantial changes for some of the lesser candidates. Furthermore, only the relative proportions of the weights matter—for example, suppose we multiply the weights by 53:

```
> lm (A2 ~ A+B+C+D+E+F+G+H+J+K+N-1, fpe, weights=53/EI) $coef
      A      B      C      D      E      F      G
1.06713 -0.10505 0.24596 0.92619 0.24940 0.75511 1.97221
      H      J      K      N
-0.56622 0.61164 1.21066 0.52935
```

This makes no difference.

Now there is one remaining difficulty, unrelated to the weighting, in that proportions are supposed to be between zero and one. We can impose an *ad hoc* fix by truncating the coefficients that violate this restriction either to zero or one as appropriate. This gives:

```
> lm (A2 ~ offset(A+G+K)+C+D+E+F+N-1, fpe, weights=1/EI) $coef
C D E F N
0.22577 0.96998 0.39020 0.74424 0.60854
```

We see that voters for the Communist candidate D apparently almost all voted for the Socialist Mitterand in the second round. However, we see that around 20% of the voters for the Gaullist candidate C voted for Mitterand. This is surprising since these voters would normally favor the more right wing candidate, Giscard. This appears to be the decisive factor. We see that of the larger blocks of smaller candidates, the Ecology party voters, E, roughly split their votes as did the first round nonvoters. The other candidates had very few voters and so their behavior is less interesting.

This analysis is somewhat crude and more sophisticated approaches are discussed in Anderson and Loynes (1987).

In cases where the form of the variance of ε is not completely known, we may model Σ using a small number of parameters. For example:

$$\text{var } \varepsilon_i = \gamma_0 + \gamma_1 x_1$$

might seem reasonable in a given situation. The iteratively reweighted least squares (IRWLS) fitting algorithm is:

1. Start with $w_i = 1$.
2. Use least squares to estimate β .
3. Use the residuals to estimate γ , perhaps by regressing $\hat{\varepsilon}^2$ on x .
4. Recompute the weights and go to 2.

Continue until convergence. There are some concerns about this because the estimation of the γ has some uncertainty and consumes some degrees of freedom. This affects the subsequent inference about β . An extensive investigation of this may be found in Carroll and Ruppert (1988).

Another approach is to model the variance and jointly estimate the regression and weighting parameters using a likelihood-based method. This can be implemented in R using the `gl` () function in the `nlme` library.

6.3 Testing for Lack of Fit

How can we tell whether a model fits the data? If the model is correct, then $\hat{\sigma}^2$ should be an unbiased estimate of σ^2 . If we have a model that is not complex enough to fit the data or simply takes the wrong form, then $\hat{\sigma}^2$ will overestimate σ^2 . The situation is illustrated in Figure 6.1. Alternatively, if our model is too complex and overfits the data, then $\hat{\sigma}^2$ will be an underestimate.

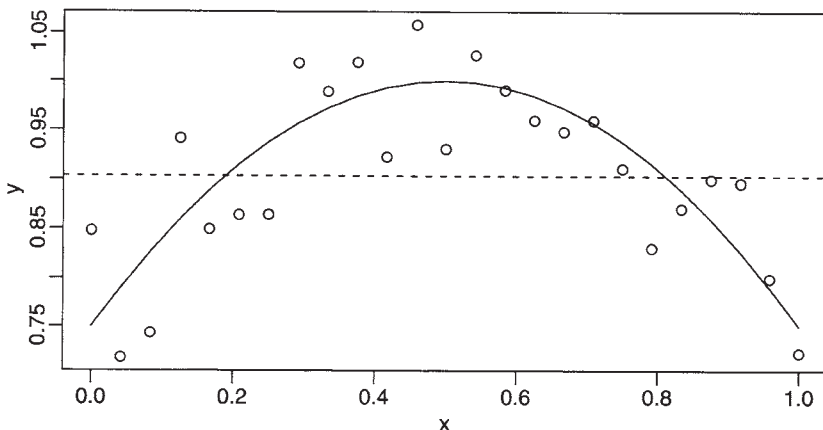


Figure 6.1 True quadratic fit shown with the solid line and incorrect linear fit shown with the dotted line. Estimate of σ^2 will be unbiased for the quadratic model, but far too large for the linear model.

This suggests a possible testing procedure—we should compare $\hat{\sigma}^2$ to σ^2 . The usual problem, of course, is that we do not know the true value of σ^2 and so the comparison cannot be made. In a few cases, we might actually know σ^2 —for example, when measurement error is the only source of variation and we know its variance because we are very familiar with the measurement device. This is rather uncommon and we do not discuss it here—see Weisberg (1985) for an example. A more realistic possibility is that we have *replication* in our data that allows an estimate of σ^2 that does not depend on any particular model.

The $\hat{\sigma}^2$ that is based in the chosen regression model needs to be compared to some model-free estimate of σ^2 . We can do this if we have repeated y for one or more fixed x . These replicates do need to be truly independent. They cannot just be repeated measurements on the same subject or unit. Such repeated measures would only reveal the within subject variability or the measurement error. We need to know the between subject variability, as this reflects the σ^2 described in the model. Let y_{ij} be the i^{th} observation in the group of replicates j .

The “pure error” estimate of σ^2 is given by SS_{pe}/df_{pe} where:

$$SS_{pe} = \sum_j \sum_i (y_{ij} - \bar{y}_j)^2$$

and degrees of freedom $df_{pe} = \sum_j (\#replicates - 1) = n - \#groups$.

If you fit a model that assigns one parameter to each group of observations with fixed x , then the $\hat{\sigma}^2$ from this model will be the pure error $\hat{\sigma}^2$. This model is just the one-way analysis of variance (ANOVA) model—see Chapter 14. Comparing this model to the regression model amounts to the lack of fit test.

The data for this example consist of 13 specimens of 90/10 Cu-Ni alloys with varying percentages of iron content. The specimens were submerged in seawater for 60 days and the weight loss due to corrosion was recorded in units of milligrams per square decimeter per day. The data come from Draper and Smith (1998). We load in and plot the data, as seen in Figure 6.2:

```
> data(corrosion)
> plot(loss ~ Fe, corrosion, xlab="nlron content",
       ylab="Weight loss")
```

We fit a straight-line model:

```
> g <- lm(loss ~ Fe, corrosion) > summary(g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    129.79      1.40     92.5  < 2e-16
```

```
Fe          -24.02          1.28      -18.8  1.1e-09 Residual
standard error: 3.06 on 11 degrees of freedom Multiple R-
Squared: 0.97, Adjusted R-squared: 0.967 F-statistic: 352
on 1 and 11 DF, p-value: 1.06e-09
```

Now show the regression line on the plot:

```
> abline (coef (g) )
```

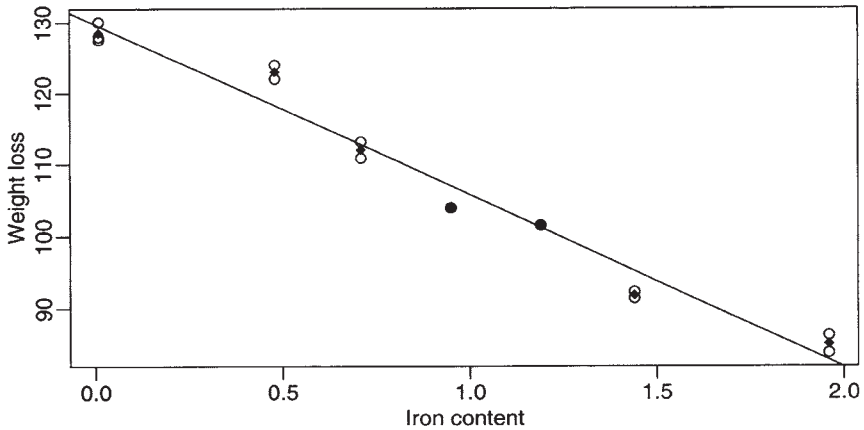


Figure 6.2 *Linear fit to the Cu-Ni corrosion data. Group means denoted by black diamonds.*

We have an R^2 of 97% and an apparently good fit to the data. We now fit a model that reserves a parameter for each group of data with the same value of x . This is accomplished by declaring the predictor to be a factor. We will describe this in more detail in chapter 14

```
> ga <- lm (loss ~ factor (Fe), corrosion)
```

The fitted values are the means in each group—put these on the plot:

```
> points (corrosion$Fe, fitted (ga), pch=18)
```

We can now compare the two models in the usual way:

```
> anova (g, ga)
Analysis of Variance Table
Model 1: loss ~ Fe
Model 2: loss ~ factor (Fe)
   Res.Df Res.Sum Sq Df Sum Sq F value Pr(>F)
1      11    102.9
2       6     11.8  5      91.1    9.28  0.0086
```

The low p-value indicates that we must conclude that there is a lack of fit. The reason is that the pure error **SD** $\sqrt{(11.8/6)} = 1.4$, is substantially less than the regression standard error of 3.06. We might investigate models other than a straight line although no obvious alternative is suggested by the plot. Before considering other models, we would first find out whether the replicates are genuine. Perhaps the low pure error SD can be explained by some correlation in the measurements. They may not be genuine replicates. Another possible explanation is that an unmeasured third variable is causing the lack of fit.

When there are replicates, it is impossible to get a perfect fit. Even when there is a parameter assigned to each group of x -values, the residual sum of squares will not be zero. For the factor model above, the R^2 is 99.7%. So even this saturated model does not attain a 100% value for R^2 . For these data, it is a small difference but in other cases, the difference can be substantial. In these cases, one should realize that the maximum R^2 that may be attained might be substantially less than 100% and so perceptions about what a good value for R^2 should be downgraded appropriately.

These methods are good for detecting lack of fit, but if the null hypothesis is accepted, we cannot conclude that we have the true model. After all, it may be that we just did not have enough data to detect the inadequacies of the model. All we can say is that the model is not contradicted by the data.

When there are no replicates, it may be possible to group the responses for similar x but this is not straightforward. It is also possible to detect lack of fit by less formal, graphical methods.

A more general question is how good a fit do you really want? By increasing the complexity of the model, it is possible to fit the data more closely. By using as many parameters as data points, we can fit the data exactly. Very little is achieved by doing this since we learn nothing beyond the data and any predictions made using such a model will tend to have a very high variance. The question of how complex a model to fit is difficult and fundamental. For example, we can fit the mean responses for the previous example exactly using a sixth order polynomial:

```
> gp <- lm (loss ~ Fe+I (Fe^2)+I (Fe^3)+I (Fe^4)+I (Fe^5)+
  I (Fe^6), corrosion)
```

Now look at this fit:

```
> plot (loss ~ Fe, data=corrosion, ylim=c (60, 130) )
> points (corrosion$Fe,fitted(ga),pch=18)
> grid <- seq(0, 2, len=50)
> lines (grid, predict (gp, data.frame (Fe=grid)))
```

as shown in Figure 6.3. The fit of this model is excellent—for example:

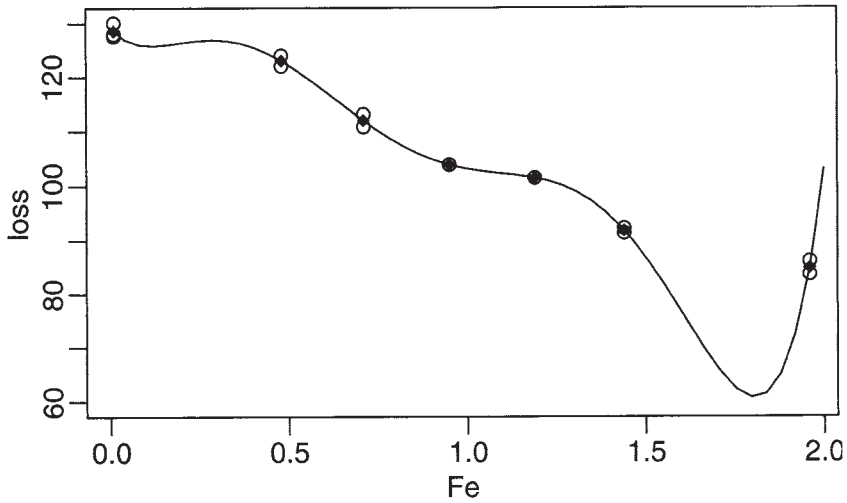


Figure 6.3 *Polynomial fit to the corrosion data.*

```
> summary(gp) $r.squared
[1] 0.99653
```

but it is clearly ridiculous. There is no plausible reason corrosion loss should suddenly drop at 1.7 and thereafter increase rapidly. This is a consequence of overfitting the data. This illustrates the need not to become too focused on measures of fit like R^2 . The fit needs to reflect knowledge of the subject matter and simplicity in modeling is a virtue.

6.4 Robust Regression

When the errors are normal, least squares regression is clearly best, but when the errors are nonnormal, other methods may be considered. Of particular concern are long-tailed error distributions. One approach is to remove the largest residuals as outliers and still use least squares, but this may not be effective when there are several large residuals because of the leave-out-one nature of the outlier tests. Furthermore, the outlier test is an accept/reject procedure that is not smooth and may not be statistically efficient for the estimation of β . Robust regression provides an alternative. There are several methods, but we will present just two popular methods.

6.4.1 M-Estimation

M-estimates choose β to minimize:

$$\sum_{i=1}^n \rho(y_i - x_i^T \beta)$$

Some possible choices for ρ are:

1. $\rho(x)=x^2$ is just least squares.
2. $\rho(x)=x$ is called least absolute deviation (LAD) regression. This is also called L_1 regression.
- 3.

$$\rho(x) = \begin{cases} x^2/2 & \text{if } |x| \leq c \\ c|x| - c^2/2 & \text{otherwise} \end{cases}$$

is called Huber's method and is a compromise between least squares and LAD regression. c should be a robust estimate of σ . A value proportional to the median of $\hat{\epsilon}$ is suitable.

Robust regression is related to WLS. The normal equations tell us that:

$$X^T(y - X\hat{\beta}) = 0$$

With weights and in nonmatrix form this becomes:

$$\sum_{i=1}^n w_i x_{ij} (y_i - \sum_{j=1}^p x_{ij} \beta_j) = 0 \quad j = 1, \dots, p$$

Now differentiating the M-estimate criterion with respect to β_j and setting to zero we get:

$$\sum_{i=1}^n \rho'(y_i - \sum_{j=1}^p x_{ij} \beta_j) x_{ij} = 0 \quad j = 1, \dots, p$$

Now let $u_i = y_i - \sum_{j=1}^p x_{ij} \beta_j$ to get:

$$\sum_{i=1}^n \frac{\rho'(u_i)}{u_i} x_{ij} (y_i - \sum_{j=1}^p x_{ij} \beta_j) = 0 \quad j = 1, \dots, p$$

so we can make the identification of $w(u) = \rho'(u)/u$. We find for our choices of ρ above that:

1. LS: $w(u)$ is constant.
2. LAD: $w(u) = 1/u$ —note the asymptote at 0—this makes a weighting approach infeasible.
3. Huber:

$$w(u) = \begin{cases} 1 & \text{if } u \leq c \\ c/u & \text{otherwise} \end{cases}$$

There are many other choices that have been used. Because the weights depend on the residuals, an IRWLS approach to fitting must be used. We can get standard errors by $\hat{\text{var}} \hat{\beta} = \hat{\sigma}^2 (X^T W X)^{-1}$ using a robust estimate of σ^2 .

We demonstrate the methods on the Galápagos Islands data. Using least squares first:

```
> data (gala)
> gl <- lm (Species ~ Area + Elevation + Nearest +
Scruz + Adjacent, gala)
> summary (gl)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.06822    19.15420   0.37    0.7154
Area          -0.02394     0.02242  -1.07    0.2963
Elevation      0.31946     0.05366   5.95   3.8e-06
Nearest        0.00914     1.05414   0.01    0.9932
Scruz          -0.24052     0.21540  -1.12    0.2752
Adjacent       -0.07480     0.01770  -4.23    0.0003
Residual standard error: 61 on 24 degrees of freedom
Multiple R-Squared:  0.766, Adjusted R-squared:  0.717
F-statistic: 15.7 on 5 and 24 DF, p-value: 6.84e-07
```

Least squares works well when there are normal errors, but performs poorly for long-tailed errors. The Huber method is the default choice of the `rlm()` function, which is part of the MASS package of Venables and Ripley (2002).

```
> library (MASS)
> gr <- ~ rlm (Species ~ Area + Elevation + Nearest +
Scruz + Adjacent, gala)
> summary (gr)
Coefficients :
              Value Std . Error t value
(Intercept)   6.36      12.390   0.513
Area          -0.006     0.015  -0.421
Elevation      0.248     0.035   7.132
Nearest        0.359     0.682   0.527
Scruz          -0.195     0.139  -1.401
Adjacent       -0.055     0.011  -4.765
Residual standard error: 29.7 on 24 degrees of freedom
```

The R^2 and F -statistics are not given because they cannot be calculated (at least not in the same way). Similarly, p -values are not given although we can use the asymptotic normality of the estimator to make approximate inferences using the t -values. The numerical values of the coefficients have changed a small amount, but the general significance of the variables remains the same and our substantive conclusion would not be altered.

We can also do LAD regression using the `quantreg` package. The default option does LAD while other options allow for quantile regression:

```
> library (quantreg)
> attach (gala)
>      qq      <      ~      rq      (Species
~Area+Elevation+Nearest+Scruz+Adjacent)
> summary (qq)
Coefficients:
              coefficients      lower bd      upper bd
(Intercept)      1.3144484     -19.877769     24.374115
Area              -0.0030600     -0.031851     0.527999
Elevation         0.2321147       0.124526     0.501955
Nearest           0.1636558     -3.163385     2.988965
Scruz             -0.1231408     -0.479867     0.134763
Adjacent          -0.0518535     -0.104577     0.017394
Degrees of freedom: 30 total; 24 residual
> detach(gala)
```

Again, there is some change in the coefficients. The confidence intervals now suggest that adjacent is not significant.

For this example, we do not see any big qualitative difference in the coefficients and for want of evidence to the contrary, we might stick with least squares as the easiest to work with. Had we seen something different, we would need to find out the cause. Perhaps some group of observations were not being fit well and the robust regression excluded these points.

6.4.2 Least Trimmed Squares

Another popular method is least trimmed squares (LTS). Here one minimizes $\sum_{i=1}^q \hat{\epsilon}_{(i)}^2$ where q is some number less than n and (i) indicates sorting. This method has a high *breakdown* point because it can tolerate a large number of outliers depending on how q is chosen. The Huber and L_1 methods will still fail if some $\epsilon_i \rightarrow \infty$ LTS is an example of a *resistant* regression method. Resistant methods are good for dealing with data where we expect there to be a certain number of bad observations that we want to have no weight in the analysis:

```
> library (lqs)
> g <- ltsreg(Species ~ Area+Elevation+Nearest+ Scruz+Adjacent,
gala)
> coef (g)
(Intercept)      Area Elevation      Nearest      Scruz
      7.410175  1.627483  0.011830  1.095214  -0.125413
      Adjacent
      -0.204292
> g <- ltsreg (Species ~ Area+Elevation+Nearest+ Scruz + Adja-
cent, gala)
```

```
> coef (g)
(Intercept)      Area Elevation  Nearest      Scruz
  5.972165  1.578810   0.025237  0.948129 -0.108829
  Adjacent
 -0.201755
```

The default choice of q is $\lfloor n/2 \rfloor + \lfloor (p+1)/2 \rfloor$ where $\lfloor x \rfloor$ indicates the largest integer less than or equal to x . I repeated the command twice and you will notice that the results are somewhat different. This is because the default genetic algorithm used to compute the coefficients is nondeterministic. An exhaustive search method can be used:

```
> g <- Itsreg (Species ~ Area + Elevation + Nearest + Scruz
+ Adjacent, gala, nsamp="exact")
> coef (g)
(Intercept)      Area Elevation  Nearest      Scruz
  9.381145  1.543658   0.024125  0.811109 -0.117732
  Adjacent
 -0.197923
```

This takes about 17 seconds on a 2.4 GHz Intel Pentium IV processor. For larger datasets, it will take much longer so this method might be impractical and the default method might be required.

This really does make substantial differences to the coefficients. For example, the Area coefficient is now substantially larger while the Elevation coefficient is substantially smaller. However, we do not have the standard errors for the LTS regression coefficients. We now use a general method for inference that is especially useful when such theory is lacking—the bootstrap.

To understand how this method works, think about how we might empirically determine the distribution of an estimator. We could repeatedly generate artificial data from the true model, compute the estimate each time and gather the results to study the distribution. This technique, called simulation, is not available to us for real data, because we do not know the true model. The bootstrap emulates the simulation procedure above except instead of sampling from the true model, it samples from the observed data. Remarkably, this technique is often effective. It sidesteps the need for theoretical calculations that may be extremely difficult or even impossible. See Efron and Tibshirani (1993) for an introductory text. To see how the bootstrap method compares with simulation, we spell out the steps involved. In both cases, we consider X fixed.

Simulation

The idea is to sample from the known distribution and compute the estimate, repeating many times to find as good an estimate of the sampling distribution of the estimator as we need. For the regression case, it is easiest to start with a sample from the error distribution since these are assumed to be independent and identically distributed:

1. Generate ε from the known error distribution.
2. Form $y = X\beta + \varepsilon$ from the known β and fixed X .
3. Compute $\hat{\beta}$.

We repeat these three steps many times. We can estimate the sampling distribution of $\hat{\beta}$ using the empirical distribution of the generated $\hat{\beta}$, which we can estimate as accurately as we please by simply running the simulation for long enough. This technique is useful for a theoretical investigation of the properties of a proposed new estimator. We can see how its performance compares to other estimators. However, it is of no value for the actual data since we do not know the true error distribution and we do not know the true β .

Bootstrap

The bootstrap method mirrors the simulation method, but uses quantities we do know. Instead of sampling from the population distribution, which we do not know in practice, we resample from the data:

1. Generate ϵ^* by sampling with replacement from $\hat{\epsilon}_1, \dots, \hat{\epsilon}_n$
2. $y^* = X\hat{\beta} + \epsilon^*$
3. $\hat{\beta}^*$ Compute from (X, y^*) .

This time, we use only quantities that we know. For very small n , it is possible to compute $\hat{\beta}^*$ for every possible sample from $\hat{\epsilon}_1, \dots, \hat{\epsilon}_n$, but usually we can only take as many samples as we have computing power available. This number of bootstrap samples can be as small as 50 if all we want is an estimate of the variance of our estimates but needs to be larger if confidence intervals are wanted.

To implement this, we need to be able to take a sample of residuals with replacement. `sample()` is good for generating random samples of indices:

```
> sample(10, rep=T)
[1] 7 9 9 2 5 7 4 1 8 9
```

and hence a random sample (with replacement) of LTS residuals is:

```
> residuals(g) [sample (30, rep=TRUE)]
   Onslow   Seymour   Onslow   Rabida Daphne.Major
-5.252656 33.841670 -5.252656 156.797891    0.040351
```

(rest deleted)

We now execute the bootstrap. We extract the fixed X and then make a matrix to save the results in. We repeat the bootstrap process 1000 times:

```
> x <- model.matrix ( ~ Area+Elevation+Nearest+Scruz+
  Adjacent, gala) [, -1]
> bcoef <- matrix (0, 1000, 6)
> for (i in 1:1000){
+   newy <- predict (g) + residuals(g)[sample(30,rep=T)]
+   brg <- ltsreg (x, newy, nsamp="best")
+   bcoef [i,] <- brg$coef
+ }
```

It is not convenient to use the `nsamp="exact"` since that would require 1000 times the time it takes to make the original estimate. That is about four hours on our computer. Being impatient, we compromised and used the second best option of `nsamp="best"`. This likely means that our bootstrap estimates of variability will be somewhat on the high side. This illustrates a common practical difficulty with the bootstrap—it can take a long time to compute. Fortunately, this problem recedes as processor speeds increase.

We can make a 95% confidence interval for this parameter by taking the empirical quantiles:

```
> quantile (bcoef [,2], c (0.025,0.975))
 2.5%  97.5%
1.4906 1.6173
```

Zero lies outside this interval so we are confident that there is an area effect. We can get a better picture of the distribution by looking at the density and marking the confidence interval:

```
> plot(density(bcoef [,2]), xlab="Coefficient of Area", main="")
> abline (v=quantile (bcoef[, 2], 0(0.025, 0.975) ) )
```

See Figure 6.4. We see that the distribution is more peaked than a normal with some longish tails.

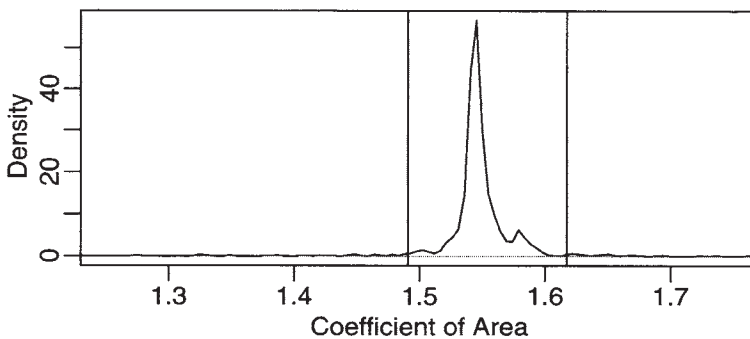


Figure 6.4 *Bootstrap distribution of $\hat{\beta}_{Area}$ with 95% confidence intervals.*

This would be more accurate if we took more than 1000 bootstrap resamples. The conclusion here would be that the area variable is significant. That is in contrast to the conclusion from the least squares fit. Which estimates are best? An examination of the Cook distances for the least squares fit shows the island of Isabela to be very influential. If we exclude this island from the least squares fit, we find that:

```
> gli <- lm (Species ~ Area + Elevation + Nearest + Scrub +
```

```

Adjacent, gala, subset= (row.names (gala) != "Isabela"))
> summary (gls)
Coefficients:
            Estimate Std. Error t value Pr(>|t|) (Intercept)
22.5861      13.4019      1.69    0.1055
Area          0.2957      0.0619    4.78  8.0e-05
Elevation      0.1404      0.0497    2.82  0.0096
Nearest       -0.2552      0.7217   -0.35  0.7269
Scruz         -0.0901      0.1498   -0.60  0.5534
Adjacent      -0.0650      0.0122  -5.32  2.1e-05
Residual standard error: 41.6 on 23 degrees of freedom
Multiple R-Squared: 0.871, Adjusted R-squared: 0.843
F-statistic: 31.2 on 5 and 23 DF, p-value: 1.62e-09

```

This fit is much closer to the LTS fit in that Area and Adjacent are very significant predictors. Thus, there are two routes to the same goal. We can use regression diagnostics in conjunction with least squares to identify bad or unusual points or we can use robust methods. The former approach is more flexible and allows for the discovery of a wider class of problems, but it is time consuming and does require human intervention. When data need to be quickly analyzed, perhaps without expert assistance or when large numbers of datasets need to be fitted, robust methods give some protection against aberrant data.

Another interesting point is that the M-estimate failed to identify the unusual island, Isabela, and gave similar results to the full data least squares fit. We can show similar behavior on another dataset—consider the Star data presented in Section 4.2.2. We compute the least squares, Huber and LTS fits and display them in Figure 6.5:

```

> data(star)
> plot (light ~ temp, star)
> gs1 <- lm (light ~ temp, star)
> abline (coef (gs1) )
> gs2 <- rlm (light ~ temp, star)
> abline (coef (gs2), lty=2)
> gs3 <- ltsreg (light ~ temp, star, nsamp="exact")
> abline (coef (gs3), lty=5)

```

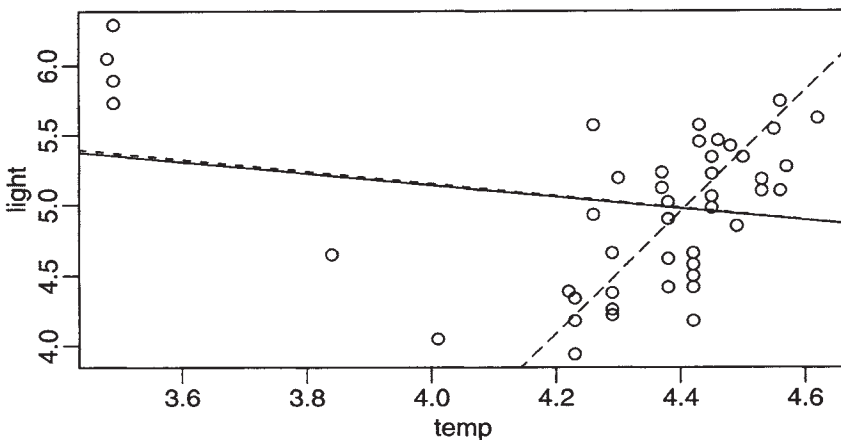


Figure 6.5 *Regression fits compared. Least squares is the solid line, Huber is the dotted line and LTS is the dashed line. Only LTS finds the fit in the bulk of the data.*

Only LTS managed to capture the trend in the main group of points. The Huber estimate is almost the same as the least squares estimate.

Summary

1. Robust estimators provide protection against long-tailed errors, but they cannot overcome problems with the choice of model and its variance structure.
2. Robust estimates just give you $\hat{\beta}$ and possibly standard errors without the associated inferential methods. Software and methodology for this inference is not easy to come by. The bootstrap is a general-purpose inferential method which is useful in these situations.
3. Robust methods can be used in addition to least squares as a confirmatory method. You have cause to worry if the two estimates are far apart. The source of the difference should be investigated.
4. Robust estimates are useful when data need to be fit automatically without the intervention of a skilled analyst.

Exercises

1. Researchers at National Institutes of Standards and Technology (NIST) collected pipeline ine data on ultrasonic measurements of the depths of defects in the Alaska pipeline in the field. The depth of the defects were then remeasured in the laboratory. These measurements were performed in six different batches. It turns out that this batch effect is not significant and so can be ignored in the analysis that follows. The laboratory measurements are more accurate than the in-field measurements, but more time consuming and expensive. We want to develop an regression equation for correcting the in-field measurements.

(a) Fit a regression model $\text{Lab} \sim \text{Field}$. Check for nonconstant variance.

(b) We wish to use weights to account for the nonconstant variance. Here we split the range of Field into 12 groups of size nine (except for the last group which has only eight values). Within each group, we compute the variance of Lab as varlab and the mean of Field as meanf field. Supposing pipeline is the name of your data frame, the following R code will make the needed computations:

```
> i <- order(pipeline$Field)
> npipe <- pipeline[i,]
> ff <- gl(12, 9)[-108]
> meanfield <- unlist(lapply(split(npipe$Field, ff),
mean))
> varlab <- unlist(lapply(split(npipe$Lab, ff), var))
```

Suppose we guess that the the variance in the response is linked to the predictor in the following way:

$$\text{var}(\text{Lab}) = a_0 \text{Field}^{a_1}$$

Regress $\log(\text{varlab})$ on $\log(\text{meanf ield})$ to estimate a_0 and a_1 . (You might choose to remove the last point.) Use this to determine appropriate weights in a WLS fit of Lab on Field. Show the regression summary.

- (c) An alternative to weighting is transformation. Find transformations on Lab and/or Field so that in the transformed scale the relationship is approximately linear with constant variance. You may restrict your choice of transformation to square root, log and inverse.
2. Using the divorce data, fit a regression model with divorce as the response and unemployed, femlab, marriage, birth and militaryas predictors.
 - (a) Make two graphical checks for correlated errors. What do you conclude?
 - (b) Allow for serial correlation with an AR(1) model for the errors. (Hint: Use maximum likelihood to estimate the parameters in the GLS fit by `glS(..., method="ML",...)`). What is the estimated correlation and is it significant? Does the GLS model change which variables are found to be significant?
 - (c) Speculate why there might be correlation in the errors.
 3. For the salmonella dataset, fit a linear model with colonies as the response and $\log(\text{dose}+1)$ as the predictor. Check for lack of fit.
 4. For the cars dataset, fit a linear model with distance as the response and speed as the predictor. Check for lack of fit.
 5. Using the stackloss data, fit a model with stack. loss as the response and the other three variables as predictors using the following methods:
 - (a) Least squares
 - (b) Least absolute deviations
 - (c) Huber method
 - (d) Least trimmed squares

Compare the results. Now use diagnostic methods to detect any outliers or influential points. Remove these points and then use least squares. Compare the results.

